

EN 508: Solid Mechanics for Energy Systems

Course Code	EN 508
Course Name	Solid Mechanics for Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solid
Intended for	UG/PG
Distribution	Compulsory for Energy Engineering (Mechanical) and Elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Introduction**
Motivation for course; Review of elementary solid mechanics. [1 Lecture]
- **Kinematics**
deformation, displacements, strain tensors, strain-displacement relations, compatibility equation. [3 Lectures]
- **Kinetics**
Traction vector, stress tensors, principle stresses, equations of motion, equilibrium. [3 Lectures]
- **Constitutive Equations**
anisotropic/orthotropic/transversely isotropic/ isotropic materials, generalized Hooke's law, linearized elasticity. [3 Lectures]
- **Basic Principles**
stress and displacement formulations for energy system, superposition, St. Venant's principle, boundary conditions. [3 Lectures]
- **Variational Methods**
strain energy, uniqueness, complementary energy, potential energy, virtual work principle, minimum total potential/complementary energy principle, approximate solutions. [4 Lectures]
- **Two-Dimensional (2-D) Theory**
plane stress and plane strain, anti-plane shear, Airy's stress function, inverse method, complex variable method, displacement solution. [3 Lectures]
- **2-D Problems**
Energy component (beam) bending, plate with a hole, curved beams, pressurized cylinders, wedges, rotating disks [3 Lectures]
- **Torsion**
St. Venant torsion theory, Prandtl stress function, membrane analogy. [3 Lectures]
- **The finite element method for numerical analyses of energy system**
Finite elements, Element, interpolation functions, Element strains, stresses and strain energy density, Element Stiffness Matrix, Global Stiffness Matrix, Boundary conditions. Demonstration of FEM based software (ANSYS) for analyzing stress, deformation and failure in components, assemblies and structures. [8 Lectures]
- **Boundary value problems for energy system made of elastic-plastic materials**
Tension-torsion of thin walled tubes, Plastic limit load, Approximate methods in metal forming. [4 Lectures]
- **Failure modes in Energy components**
Fracture, Fatigue, Buckling, Large deflections, Plastic collapse. [4 Lectures]

Text and Reference Books

1. S. P. Timoshenko and J. N. Goodier, **Theory of Elasticity**, 3rd Edition, McGraw-Hill, New York, 1970
2. W. S. Slaughter, **The Linearized Theory of Elasticity**, Birkhauser, 2002 .
3. R. W. Little, **Elasticity**, Prentice-Hall, 1973.
4. P. Boresi and K. P. Chong, **Elasticity in Engineering Mechanics**, 2nd Edition, Wiley, 2000.
5. Hertzberg R.W., Vinci R.P., Hertzberg J., **Deformation and Fracture Mechanics of Engineering Materials**, Wiley, 2012.